Multi-trait selection of promising quinoa (*Chenopodium quinoa* Willd.) accessions evaluated in Ayacucho, Peru

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DECLARATIONS & STATEMENTS

Acknowledgments

The authors acknowledge…

Funding

Data and code availability

All original contributions generated and analyzed in this study are included in the article and its supplementary materials. The reproducible datasets and analytical workflows are provided in Supplementary File 1 and are accessible through the GitHub repository at:

Author contributions

The conception and design of the study were undertaken by F.G.F. and F.L.I. Data acquisition, curation, formal analysis, and software implementation were performed by S.C.N., M.N.M.S., and F.L.I. Field and laboratory investigation, as well as validation of experimental procedures, were conducted by F.G.F. and W.M.M., who also contributed to the critical revision of the manuscript. Methodology development was carried out by F.G.F. and F.L.I., while project supervision and administration were the responsibility of F.L.I. Funding acquisition and the provision of essential resources were ensured by M.O.C. Data visualization was produced by S.C.N., M.N.M.S., and F.L.I. The first draft of the manuscript was written by F.G.F., M.N.M.S., and F.L.I., and all authors contributed to the critical review and editing of subsequent versions. All authors read and approved the final manuscript.

Conflict of Interest

The authors declare that they have no conflicts of interest.

ABSTRACT

Keywords:

INTRODUCTION

Quinoa (*Chenopodium quinoa* Willd.) is an important plant genetic resource for agrobiodiversity conservation and crop improvement because of its phenotypic variation, nutritional value, and capacity to grow under contrasting agroecological conditions (Manjarres-Hernández et al., 2021; Ain et al., 2023; Nepal et al., 2023). This variation offers opportunities to identify accessions with desirable agronomic traits. However, its effective use requires detailed accession characterization and selection criteria that reflect specific breeding objectives (Lozano-Isla et al., 2023; Baldera-Chapoñan et al., 2025). Agromorphological characterization allows differences in plant architecture, phenology, inflorescence traits, and yield components to be documented, providing a phenotypic basis for germplasm evaluation and prioritization (Stanschewski et al., 2021; Craine et al., 2023).

Quinoa diversity is evident among accessions associated with different growing environments and selection processes. Based on geographical distribution and the agroclimatic conditions of growing areas, five main ecotypes are recognized: inter-Andean valleys, northern and southern Altiplano in the Andean region, sea-level areas along the coast, and Yungas on the eastern slopes of the Andes. These ecotypes reflect the influence of altitude, temperature, and water availability on crop adaptation (Del Pozo et al., 2023; Vásquez et al., 2024). While this classification helps describe the ecogeographical diversity of quinoa, it does not directly determine the agronomic performance of individual accessions. Within a germplasm collection, accessions may vary in plant height, branching, panicle dimensions, grain size, and yield, depending on their genetic constitution and the environment in which they are evaluated (Ain et al., 2023; De La Cruz et al., 2025; Matías et al., 2023). These traits can be used to assess phenotypic variability and identify combinations of characteristics for genetic improvement (Craine et al., 2023).

Qualitative and quantitative traits offer complementary information for assessing diversity within germplasm collections. Qualitative traits describe morphological categories, including leaf shape, organ pigmentation, and panicle architecture, while quantitative traits measure characteristics such as plant height, reproductive organ dimensions, thousand-seed weight, and yield. Considering both types of traits provides a clearer picture of phenotypic variation among accessions and reveals which traits show the most distinct differences. (Delgado et al., 2024; Baldera-Chapoñan et al., 2025; Checmapocco-Conza et al., 2026). However, phenotypic diversity alone does not determine which accessions are suitable for a given breeding objective, as favorable performance in one trait may be accompanied by less desirable values in others. Multi-trait selection addresses this challenge by combining several attributes into a common criterion for accession prioritization (Ahmed et al., 2023).

Multi-trait selection methods differ in their statistical foundations and how they incorporate agronomic objectives. Principal component analysis (PCA) transforms the original variables into orthogonal components that explain decreasing proportions of the total variance, allowing the multivariate structure of the dataset to be examined (Jolliffe & Cadima, 2016). The Elston index identifies accessions that satisfy predefined thresholds for the target traits (Elston, 1963), while the Mulamba–Mock index combines trait rankings by summing the ranks assigned to each accession (Mulamba & Mock, 1978). Ideotype distance measures the difference between an accession's observed trait profile and a predefined target profile (Crossa et al., 2017). Random Forest uses ensembles of decision trees to model relationships between predictor variables and a response variable. Applying these methods to accession selection requires defining the prediction target and establishing how the predicted values will be used to rank accessions (Breiman, 2001). Given these methodological differences, the same collection of accessions may produce different rankings even when identical traits are considered.

Comparing multi-trait selection methods allows relationships among selection scores, similarities between rankings, and overlap among selected accessions to be examined. Such comparisons help identify accessions that are consistently prioritized across methods and those whose rankings depend on the analytical criteria used. In Ayacucho, the evaluation of quinoa accessions originating from different regions of Peru offers an opportunity to examine how these methods influence germplasm prioritization under the experimental conditions. Agreement among methods may indicate potentially promising candidates, while differences in rankings can reveal accessions associated with specific phenotypic objectives. Both outcomes can inform the initial prioritization of germplasm, although agronomic superiority must be confirmed through experimental validation.

Although different multi-trait selection methods have been applied in germplasm evaluation, it remains unclear whether these approaches produce consistent rankings and identify the same promising quinoa accessions when applied to a common dataset. This uncertainty highlights the need to compare their performance and assess the agreement among selected accessions under the same experimental conditions. This study aimed to: (i) characterize the phenotypic diversity of 144 quinoa accessions using qualitative and quantitative traits; and (ii) compare five multi-trait selection methods, namely PCA, the Elston index, the Mulamba–Mock index, ideotype distance, and Random Forest, to identify promising accessions and assess agreement among the resulting rankings.

MATERIALS AND METHODS

Study area

The field experiment was conducted at the Canaán Agricultural Experimental Station of the National Institute for Agrarian Innovation (INIA), located in the district of Andrés Avelino Cáceres Dorregaray, Huamanga Province, Ayacucho Department, Peru (13°10′02″ S, 74°12′26″ W), at an elevation of 2,735 m above sea level (Figure 1). The crop was evaluated during the 2022–2023 growing season, from December 2022 to August 2023. During this period, the mean temperature was 17.7 °C, with minimum and maximum temperatures of 9.9 and 24.8 °C, respectively, while annual precipitation reached 475.27 mm. The experimental soil had a loam texture, with a pH of 7.7, electrical conductivity of 14.9 dS m^{-1}, organic matter content of 1.66%, total nitrogen content of 0.08%, available phosphorus content of 25.2 mg kg^{-1}, and potassium content of 683.7 mg kg^{-1}. The cation exchange capacity was 15.74 cmol(+) kg^{-1} (Supplementary Table 1).

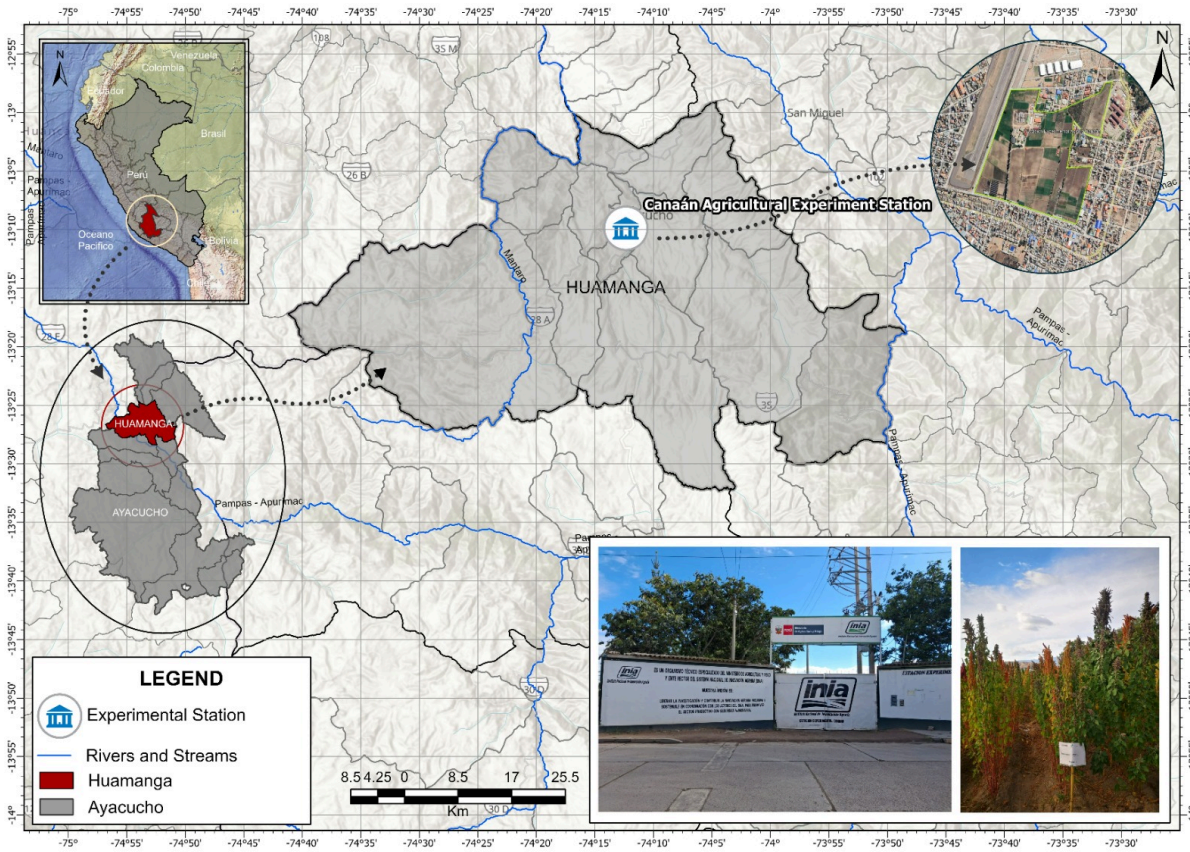


Figure 1. Location of the experimental site at the Canaán Agricultural Experimental Station (INIA), Andrés Avelino Cáceres Dorregaray district, Huamanga Province, Ayacucho, Peru.

Biological material

A total of 144 quinoa accessions maintained in the Germplasm Bank of the Laboratory of Plant Genetics and Biotechnology, Professional School of Agronomy, National University of San Cristóbal de Huamanga (UNSCH), Ayacucho, Peru, were evaluated. These accessions were collected from Ayacucho, Cusco, Apurímac, and Huancayo. Details on the identification and geographical origin of each accession are provided in Supplementary Table S2.

Field trials and management

The 144 accessions were evaluated in a 12 × 12 simple lattice design with two replications. Each experimental plot measured 3.0 × 2.4 m and consisted of three rows spaced 0.80 m apart. Seeds were sown at a rate of 10 kg ha^{-1}, with a plant density of 20 plants per linear meter.

Land preparation included plowing, harrowing, and mechanized furrow formation. Fertilizers were applied at rates of 80 kg ha^{-1} of nitrogen (N), 80 kg ha^{-1} of phosphorus pentoxide (P_2O_5), and 40 kg ha^{-1} of potassium oxide (K_2O), using urea, diammonium phosphate, and potassium chloride as fertilizer sources. Carbendazim- and mancozeb-based fungicides were applied to control downy mildew caused by *Peronospora variabilis*. Plants were harvested at physiological maturity, and the harvested material was dried in a drying tunnel, threshed, and cleaned to obtain the grain samples for evaluation.

Morphological characterization

Morphological characterization was performed following the descriptors for quinoa (*Chenopodium quinoa* Willd.) and its wild relatives established by Bioversity International et al. (2013). A total of 41 descriptors were recorded at the vegetative growth, flowering, and physiological maturity stages and grouped into qualitative or quantitative traits (Supplementary Table S3).

Qualitative descriptors included plant growth density (PLD), growth type (GT), growth habit (GH), main stem shape (SMS), main stem color (MSC), presence of pigmented axils (PPA), presence of stem striations (PST), striation color (STC), presence of branching (PB), primary branch position (PPB), leaf shape (LS), leaf margin (LM), petiole color (PC), leaf blade color (LLC), leaf granule color (LGC), male sterility (MST), panicle color at flowering (PCF) and physiological maturity (PCPM), panicle shape (PS), panicle density (PD), degree of shattering (DH), perigonium appearance (PA), perigonium color (PGC), pericarp appearance (PCA), pericarp color (PCC), episperm appearance (EA), episperm color (EC), and grain shape (GS).

Quantitative descriptors included percentage of lodged plants (LOD), number of primary branches (NPB), number of leaf teeth (NLT), petiole length (PTL), maximum leaf length (MLL), maximum leaf width (MLW), grain yield per plant (SYP), plant height (PH), stem diameter (SD), panicle length (PLr), panicle diameter (PDr), grain diameter (GD), and thousand-seed weight (TSW). To reduce border effects, observations were taken from plants in the central row of each experimental unit. Ten plants per experimental unit were evaluated for individual plant measurements.

Statistical analysis

All Statistical analyses and the preparation of figures and tables were performed in R v4.6.1 (R Core Team, 2026). Diversity among qualitative traits was estimated using 28 descriptors, considering the number of accessions in each class (Na), the number of observed classes (Nc), and their relative frequencies (fi). These frequencies were used to calculate the effective number of classes (Ne), Nei's diversity index (He; Nei, 1987) (He), and the Shannon–Weaver diversity index (H′; Shannon & Weaver, 1949). Standard errors and 95% confidence intervals were estimated for each index using 2,000 bootstrap resamples, and mean values were calculated across the 28 descriptors.

The 13 quantitative traits were analyzed using linear mixed-effects models implemented through the *H2cal()* function in the inti package v0.7.5.9 (Lozano-Isla et al., 2026), which relies on lme4 (Bates et al., 2015). The following model was fitted separately for each trait:

$$y \sim 1 + (1|\text{block}) + (1|\text{sub} - \text{block}) + (1|\text{geno})$$

where y represents the observed trait value, and block, sub-block, and geno represent the random effects of block, sub-block, and genotype, respectively. Variance components were used to estimate repeatability, while the best linear unbiased estimates (BLUEs) for each accession were used to calculate the mean, minimum and maximum values, and coefficient of variation for each trait.

Five multi-trait selection methods were applied to the BLUEs of grain yield (SYP), thousand-seed weight (TSW), plant height (PH), and stem diameter (SD), after normalizing the values to a scale of 0 to 1. The 14 highest-ranked accessions, corresponding to approximately 10% of the 144 accessions evaluated, were selected. The methods included principal component analysis (PCA), performed using PCA() from FactoMineR v2.17 (Lê et al., 2008); the Elston index, calculated using elston() from st4gi v2.2.2 (Eyzaguirre & Okola, 2026); the Mulamba–Mock index, calculated using rank() from base R; weighted Euclidean distance to the ideotype; and Random Forest, implemented using ranger() with out-of-bag (OOB) predictions. Pearson's correlation coefficients were calculated using cor() from base R to assess relationships among methods and visualized using pairs.panels() from psych v2.6.9 (William Revelle, 2026). Overlap among selected accessions was examined through set intersection, while selection differentials were calculated as standardized differences in means. Consensus was defined as an accession being selected by at least three methods.

RESULTS

Climatic conditions

Monthly precipitation peaked in March 2023 at approximately 160 mm, followed by a gradual decline to nearly zero between June and November. During the period from December 2022 to November 2023, maximum temperatures ranged from 22 to 27 °C, mean temperatures from 16 to 19 °C, and minimum temperatures from 6 to 12 °C (Figure 2) .

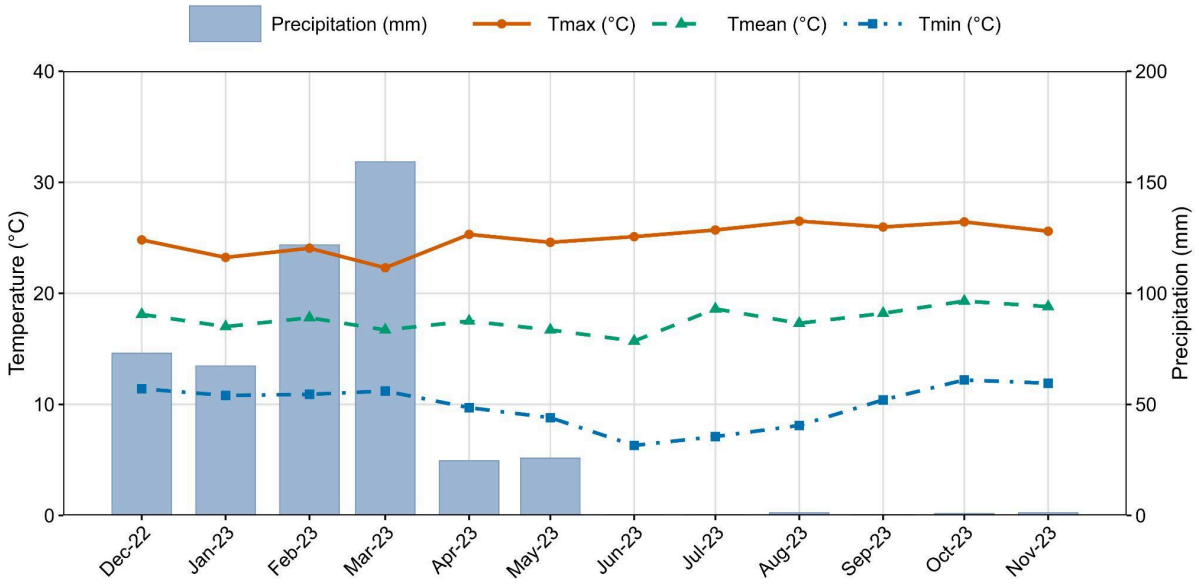


Figure 2. Monthly precipitation and maximum, mean, and minimum air temperatures recorded at the Canaán Agricultural Experimental Station from December 2022 to November 2023.

Agromorphological variability of quinoa germplasm

Phenotypic diversity of qualitative traits

To identify the morphological traits showing the greatest variation among quinoa accessions, phenotypic diversity was assessed using 28 qualitative descriptors. The mean diversity indices were 0.49 ± 0.04 for Nei's index (He) and 0.89 ± 0.10 for the Shannon–Weaver index (H′). The highest values for both indices were observed for panicle color at physiological maturity (He = 0.86 ± 0.01; H′ = 2.14 ± 0.06), followed by perigonium color (He = 0.84 ± 0.02; H′ = 2.12 ± 0.08) and pericarp color (He = 0.75 ± 0.03; H′ = 1.72 ± 0.08) (Table 1). Yellow was the predominant panicle color at physiological maturity, while cream and pink were the most frequent perigonium colors, and cream predominated in the pericarp (Figure 3, Supplementary Table 4).

Episperm color also showed relatively high diversity (He = 0.69 ± 0.03; H′ = 1.49 ± 0.08), with white being the most frequent color. Similar values were recorded for main stem color (He = 0.69 ± 0.03; H′ = 1.44 ± 0.07), with yellow predominating (Supplementary Table 4). Among traits not associated with coloration, plant growth density had the highest Nei's diversity index (He = 0.65 ± 0.01). In contrast, the lowest diversity was observed for the presence of stem striations (He = 0.03 ± 0.02; H′ = 0.07 ± 0.04) and the presence of branching (He = 0.07 ± 0.03; H′ = 0.15 ± 0.05) (Table 1).



Figure 3. Variation in seed color among quinoa accessions in the germplasm collection.

Table 1. Phenotypic diversity indices for 28 qualitative traits in the quinoa germplasm collection.

Qualitative trait	Nc	Ne	He ± SE	He 95% CI	H′ ± SE	H′ 95% CI
Planting density	3	2.84	0.65 ± 0.01	0.61–0.66	1.07 ± 0.02	1.02–1.09
Growth type	2	1.96	0.49 ± 0.01	0.46–0.50	0.68 ± 0.01	0.65–0.69
Growth habit	4	1.46	0.32 ± 0.05	0.22–0.40	0.60 ± 0.07	0.43–0.73
Main stem shape	2	1.98	0.49 ± 0.01	0.47–0.50	0.69 ± 0.01	0.66–0.69
Main stem color	7	3.18	0.69 ± 0.03	0.62–0.74	1.44 ± 0.07	1.27–1.55
Presence of pigmented axils	3	1.86	0.46 ± 0.04	0.38–0.53	0.78 ± 0.06	0.65–0.87
Presence of stem striations	2	1.03	0.03 ± 0.02	0.00–0.07	0.07 ± 0.04	0.00–0.15
Striation color	4	2.26	0.56 ± 0.02	0.51–0.59	0.91 ± 0.05	0.80–1.00
Presence of branching	2	1.07	0.07 ± 0.03	0.01–0.13	0.15 ± 0.05	0.04–0.25
Primary branch position	2	1.18	0.15 ± 0.04	0.08–0.23	0.29 ± 0.06	0.17–0.39
Leaf shape	2	1.91	0.48 ± 0.02	0.42–0.50	0.67 ± 0.02	0.62–0.69
Leaf margin	3	1.68	0.40 ± 0.04	0.31–0.49	0.73 ± 0.06	0.59–0.85
Petiole color	3	1.6	0.37 ± 0.04	0.28–0.45	0.67 ± 0.07	0.53–0.79
Leaf blade color	2	1.56	0.36 ± 0.04	0.28–0.43	0.55 ± 0.04	0.45–0.62
Leaf granule color	4	2.9	0.65 ± 0.02	0.61–0.68	1.14 ± 0.04	1.04–1.21
Presence of male sterility	2	1.64	0.39 ± 0.04	0.31–0.45	0.58 ± 0.04	0.49–0.64
Panicle color at flowering	4	2.51	0.60 ± 0.02	0.55–0.64	1.05 ± 0.06	0.93–1.14
Panicle color at physiological maturity	12	6.93	0.86 ± 0.01	0.82–0.87	2.14 ± 0.06	1.96–2.21
Panicle shape	3	1.92	0.48 ± 0.04	0.39–0.55	0.84 ± 0.06	0.71–0.94
Panicle density	3	2.69	0.63 ± 0.02	0.58–0.65	1.04 ± 0.03	0.97–1.08

Degree of dehiscence (shattering)	3	2.39	0.58 ± 0.02	0.52–0.62	0.96 ± 0.04	0.86–1.02
Perigonium appearance	2	1.65	0.39 ± 0.03	0.32–0.45	0.58 ± 0.04	0.50–0.64
Perigonium color	14	6.21	0.84 ± 0.02	0.80–0.86	2.12 ± 0.08	1.92–2.21
Pericarp appearance	2	1.53	0.35 ± 0.04	0.27–0.41	0.53 ± 0.04	0.44–0.60
Pericarp color	11	4.01	0.75 ± 0.03	0.69–0.79	1.72 ± 0.08	1.52–1.83
Episperm appearance	2	1.55	0.35 ± 0.04	0.28–0.42	0.54 ± 0.04	0.45–0.61
Episperm color	8	3.18	0.69 ± 0.03	0.62–0.73	1.49 ± 0.08	1.30–1.61
Grain shape	3	2.34	0.57 ± 0.02	0.52–0.61	0.94 ± 0.04	0.84–1.00
Mean ± SE	4.07 ± 0.62	2.39 ± 0.26	0.49 ± 0.04	0.41–0.57	0.89 ± 0.10	0.69–1.09

Note. Nc = number of observed classes; Ne = effective number of classes; He = Nei's diversity index; H′ = Shannon–Weaver diversity index; SE = standard error; CI = confidence interval. Standard errors and 95% confidence intervals were estimated using 2,000 bootstrap resamples.

Variability of quantitative traits

To characterize differences in growth, morphology, and yield among the 144 quinoa accessions, variation was assessed across 13 quantitative traits. The highest coefficients of variation were recorded for the percentage of lodged plants (CV = 104.26%) and grain yield per plant (CV = 92.04%). Lodging ranged from 0 to 68.37%, with a mean of 11.43%, while grain yield per plant ranged from 2.85 to 146.94 g, with a mean of 26.33 g. Considerable variation was also observed in stem diameter (mean = 11.42 mm; range = 4.20–26.56 mm), plant height (mean = 124.69 cm; range = 58.51–197.62 cm), and panicle length (mean = 41.69 cm; range = 13.45–80.45 cm). Among grain-related traits, thousand-seed weight averaged 2.81 g (range = 1.50–4.39 g), whereas grain diameter showed the lowest coefficient of variation (CV = 7.42%), with a mean of 1.93 mm (range = 1.57–2.24 mm) (Table 2).

Repeatability estimates ranged from 0.81 for the percentage of lodged plants to 0.99 for the number of leaf teeth. High repeatability was also recorded for maximum leaf width and thousand-seed weight (0.98 each), grain diameter (0.96), and plant height (0.95). Grain yield per plant, stem diameter, and panicle length showed repeatability estimates of 0.83, 0.93, and 0.84, respectively (Table 2).

Table 2. Descriptive statistics, coefficients of variation, and repeatability estimates for 13 quantitative traits evaluated in the quinoa germplasm collection.

Quantitative trait	Mean	Min	Max	CV(%)	Repeatability
Percentage of lodged plants (%)	11.43	0	68.37	104.26	0.81
Number of primary branches	13.61	4.68	32.66	37.19	0.85
Number of leaf teeth	9.56	1.09	23.64	52.4	0.99
Petiole length (cm)	4.27	2.03	7.04	25.86	0.95
Maximum leaf length (cm)	6.32	3.66	9.5	21.8	0.94
Maximum leaf width (cm)	5.37	1.59	9.68	30.12	0.98
Seed yield per plant (g/plant)	26.33	2.85	146.94	92.04	0.83
Plant height (cm)	124.69	58.51	197.62	24.48	0.95
Stem diameter (mm)	11.42	4.2	26.56	40.19	0.93
Panicle length (cm)	41.69	13.45	80.45	27.33	0.84
Panicle diameter (cm)	10.65	3.25	25.7	52.44	0.86
Grain diameter (mm)	1.93	1.57	2.24	7.42	0.96
Thousand-seed weight (g)	2.81	1.5	4.39	20.67	0.98

Note. Mean = mean value; Min = minimum value; Max = maximum value; CV = coefficient of variation.

Comparison of multi-trait selection methods

The five selection methods were compared to determine whether they prioritized the same quinoa accessions and favored similar combinations of agronomic traits. The greatest overlap occurred between PCA and ideotype distance, with nine accessions selected by both methods. PCA and Random Forest shared seven accessions, while the other method combinations had between one and eight accessions in common (Figure 4 A).

Selection differentials differed across methods and traits (Figure 4 B). Ideotype distance produced the highest differential for grain yield per plant (SYP; 2.22), while PCA showed the highest values for plant height (PH; 1.55) and stem diameter (SD; 1.75), along with a differential of 1.84 for SYP. The Mulamba–Mock index yielded the highest differential for thousand-seed weight (TSW; 1.14), compared with 0.27 for PCA and 0.43 for ideotype distance. Random Forest showed positive differentials for SYP (1.31), PH (1.48), and SD (1.36), but a negative differential for TSW (−0.09). Similarly, the Elston index produced positive differentials for SYP (1.73), PH (0.63), and SD (0.70), while TSW showed a negative value (−0.10).

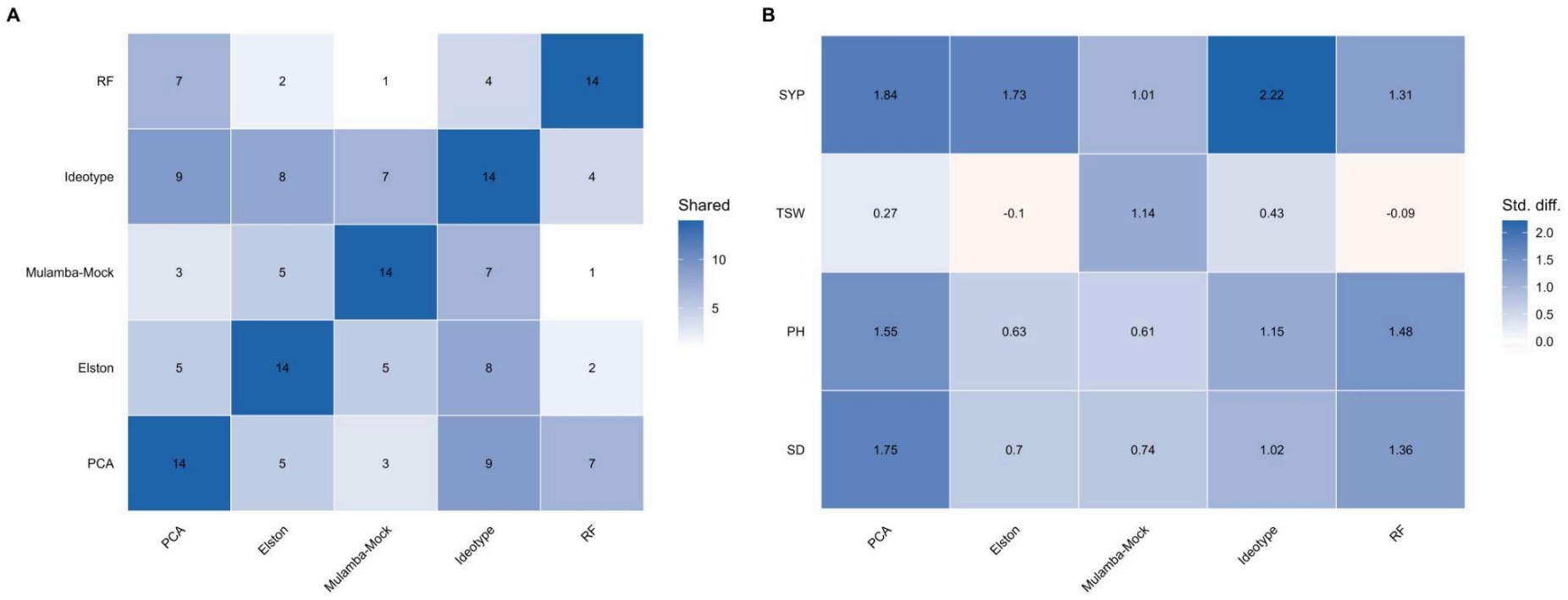


Figure 4. Comparison of five multi-trait selection methods. (A) Overlap among the 14 accessions selected by each method. (B) Standardized selection differentials for grain yield per plant (SYP), plant height (PH), stem diameter (SD), and thousand-seed weight (TSW).

Twelve accessions were selected by at least three of the five methods and were classified as consensus accessions (Table 3). Of these, UNSCH-126-2013, UNSCH-287-2022, and UNSCH-83-2013 were selected by four methods. The remaining nine accessions (UNSCH-1-2013, UNSCH-10-2013, UNSCH-125-2013, UNSCH-129-2013, UNSCH-25-2013, UNSCH-56-2013, UNSCH-57-2013, UNSCH-61-2013, and UNSCH-72-2013) were identified by three methods. Ideotype distance selected 11 of the 12 consensus accessions, followed by PCA with nine.

Grain yield among the consensus accessions ranged from 45.64 g per plant in UNSCH-10-2013 to 146.94 g per plant in UNSCH-61-2013. Plant height ranged from 115.19 cm in UNSCH-1-2013 to 191.56 cm in UNSCH-61-2013, and TSW from 2.40 g in UNSCH-72-2013 to 4.05 g in UNSCH-125-2013. Three accessions (UNSCH-1-2013, UNSCH-10-2013, and UNSCH-125-2013) had TSW values above 3.00 g but grain yields below 55 g per plant. By comparison, UNSCH-56-2013, UNSCH-57-2013, and UNSCH-61-2013 had grain yields between 80.78 and 146.94 g per plant, with TSW values ranging from 2.72 to 2.86 g. UNSCH-61-2013 had the highest grain yield in the collection, while UNSCH-287-2022 recorded the greatest panicle length.

The 14 accessions listed also differed in their qualitative traits (Table 3). Pericarp color was cream in six accessions (43%), golden yellow in five (36%), yellow in two (14%), and light brown in one (7%; UNSCH-287-2022) (Supplementary Figure 1a). Panicle density was intermediate or compact in five accessions each (36%), and lax in four (29%) (Supplementary Figure 1b). Nine accessions (64%) had a simple growth habit, four (29%; UNSCH-1-2013, UNSCH-10-2013, UNSCH-56-2013, and UNSCH-287-2022) showed branching up to the second third, and one (7%; UNSCH-25-2013) had an undefined main panicle (Supplementary Figure 1c). Glomerulate panicles were recorded in nine accessions (64%), amaranthiform panicles in four (29%; UNSCH-25-2013, UNSCH-61-2013, UNSCH-70-2013, and UNSCH-218-2015), and an intermediate panicle shape in one (7%; UNSCH-1-2013) (Supplementary Figure 1d). UNSCH-61-2013, the accession with the highest grain yield, had a golden-yellow pericarp, compact panicle density, simple growth habit, and amaranthiform panicle shape (Supplementary Table S5).

Table 3. Phenotypic characteristics of selected quinoa accessions in the germplasm collection.

Accession	N	PCA	Elston	Mulamba-Mock	Ideotype	RF	SYP	TSW	PH	SD
UNSCH-126-2013	4	✓	✓	✓	✓		90.81	3.14	141.07	16.9
UNSCH-287-2022	4	✓	✓	✓	✓		108.87	2.87	148.86	13.14
UNSCH-83-2013	4	✓	✓		✓	✓	71.92	2.79	162.65	17.11
UNSCH-1-2013	3		✓	✓	✓		49.05	3.49	115.19	12.74
UNSCH-10-2013	3		✓	✓	✓		45.64	3.67	142.31	12.39
UNSCH-125-2013	3	✓		✓	✓		54.07	4.05	160.19	21.55
UNSCH-129-2013	3		✓	✓	✓		61.8	2.94	150.23	16.03
UNSCH-25-2013	3	✓	✓		✓		83.99	2.47	169.42	19.06
UNSCH-56-2013	3	✓			✓	✓	100	2.72	188.48	16.92
UNSCH-57-2013	3	✓			✓	✓	80.78	2.86	172.77	19.58
UNSCH-61-2013	3	✓			✓	✓	146.94	2.72	191.56	21.32
UNSCH-72-2013	3	✓	✓			✓	71.23	2.4	167.13	18.41
UNSCH-218-2015	2			✓	✓		55.55	3.4	170.31	13.13
UNSCH-278-2022	2		✓		✓		110.05	2.61	147.85	9.5

Note. N indicates the number of methods that selected each accession. The methods were the Elston index, the Mulamba–Mock index, principal component analysis (PCA), ideotype distance, and Random Forest (RF). SYP: grain yield; TSW: thousand-seed weight; PH: plant height; PLr: panicle length. Accessions selected by at least three methods were considered consensus accessions.

Pearson correlations among the five selection methods were positive and significant ($p < 0.001$), with coefficients ranging from 0.37 to 0.91 (Figure 5). Mulamba–Mock and ideotype distance had the strongest correlation (r = 0.91), followed by PCA and Random Forest (r = 0.87), PCA and ideotype distance (r = 0.86), and Elston and ideotype distance (r = 0.81). The lowest correlations were found between Elston and Random Forest (r = 0.37), Mulamba–Mock and Random Forest (r = 0.55), and PCA and Elston (r = 0.60). Elston and Random Forest scores had more positively skewed distributions, while Mulamba–Mock scores were spread over a wider range (Figure 5).

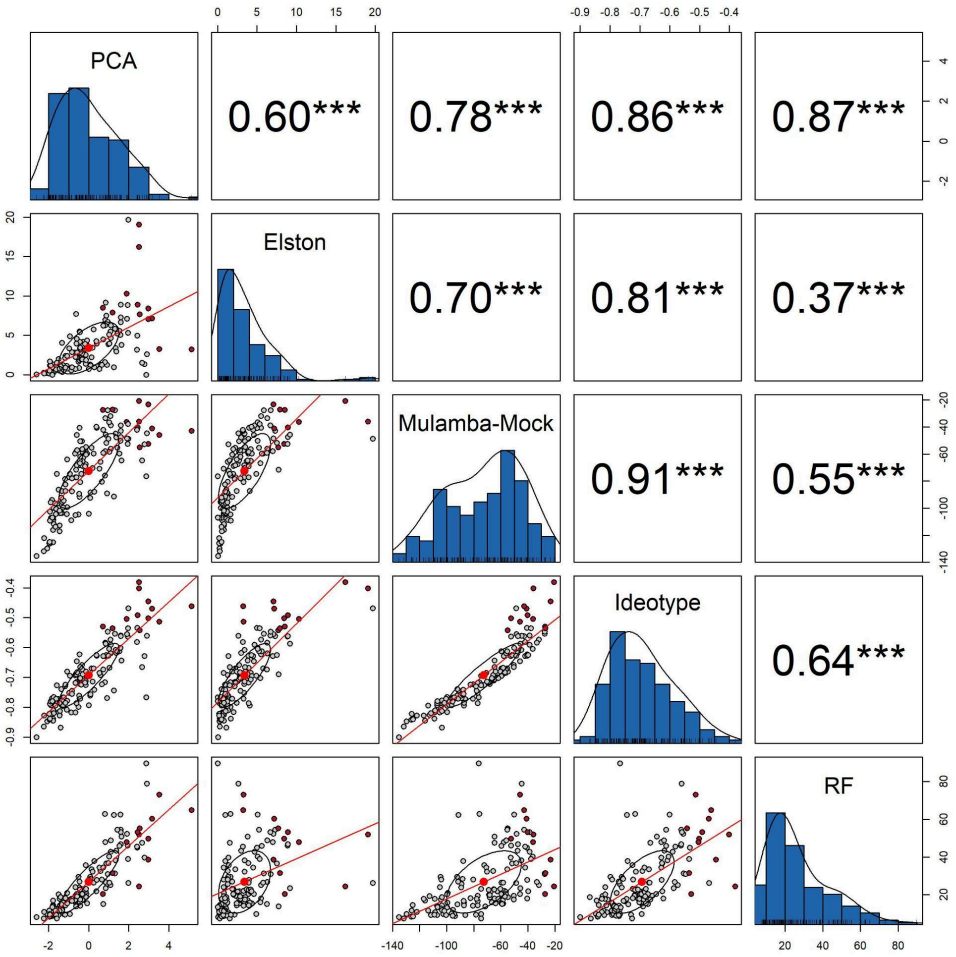


Figure 5. Distribution of selection scores and Pearson correlations among five multi-trait selection methods: principal component analysis (PCA), Elston index, Mulamba–Mock index, ideotype distance, and Random Forest (RF). Histograms represent score distributions, scatterplots show pairwise relationships, and coefficients in the upper triangle indicate Pearson correlations. Red points identify consensus accessions. Asterisks indicate statistical significance ($p < 0.001$).

DISCUSSION

⇒ **Research summary**
⇒ **Title based on the objectives (*)**
- **Main results with cross references**
- **Comparison with other works**
- **Discuss finding explanation**

⇒ **Limitations and perspectives (*)**

Research summary

Specific Objective 1

Specific Objective 2

Limitations and perspectives

CONCLUSIONS

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